

Classification of Waste Materials using CNN Based on Transfer Learning

Sujan Poudel

Department of Computer Science & Engineering
Kathmandu University
Dhulikhel, Nepal
poudelsujan03@gmail.com

Prakash Poudyal *

Department of Computer Science & Engineering
Kathmandu University
Dhulikhel, Nepal
prakash@ku.edu.np

ABSTRACT

Waste Management is important for humans as well as nature for healthy life and a clean environment. The major step for effective waste management is the segregation of waste according to its types. The advancement of technology such as hardware and artificial intelligence is used for the segregation of waste. There are several machine learning and deep learning algorithms available for image classification. Among them, Convolutional Neural Network is the most used one. The main objective of this work is to classify images of waste materials using CNN into seven categories (cardboard, glass, metal, organic, paper, plastic, and trash). Then, cardboard, organic, and paper class images are considered biodegradable waste, and other classes are considered non-biodegradable waste. The pre-trained CNN model such as InceptionV3, InceptionResNetV2, Xception, VGG19, MobileNet, ResNet50 and DenseNet201 have been trained and performed fine-tuning on the waste dataset. Among these models, the VGG19 model performed with less accuracy, whereas the InceptionV3 model performed with high learning accuracy. Overall, the obtained result is promising.

KEYWORDS

Waste Management, Waste types, Convolution Neural Network (CNN), Transfer learning, Fine-tuning

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1 INTRODUCTION

Waste Management is one of the biggest challenges in developing countries. Waste classification is a key step and it plays important role in waste management. Waste classification means a process of categorizing or segregating similar kinds of waste. According to [5] house is the main source of generating waste in Nepal. Out of total waste, household waste covers 50 to 75%. The effective segregation of waste in the house leads to a decrease in waste

in landfill sites since household waste mainly consists of organic or degradable waste other than non-degradable waste. Organic or biodegradable waste mainly consists of paper, textile, and agricultural waste whereas non-biodegradable waste is comprised of plastic, glass, metals, trash, and so on. There are other various types of waste like electrical, hazardous, and medical waste that were not included in either of these two categories in this work.

Manually, the separation of waste is hazardous for human health as well as time-consuming and haphazard. Therefore, the involvement of technology such as the Internet of Things (IoT) and Artificial Intelligence (AI) can address this problem. The distinct ability of AI to classify images leads to a successful classification of waste types. The optimal utilization of waste is necessary for a sustainable society. Reduce, Reuse, and Recycle are the 3Rs principle of waste management [1]. Thus, the primary concern about waste management should be focused on using technology to reduce waste as much as possible.

The secondary concern of waste management is the proper classification of waste as biodegradable and non-biodegradable. Thus, we purpose to classify a different class of waste and provide information about whether it is biodegradable or non-biodegradable. Initially, waste images have been classified into seven types (cardboard, glass, metal, organic, paper, plastic, and trash) using transfer learning techniques in CNN-based pre-trained models such as InceptionV3, InceptionResNetV2, Xception, VGG19, MobileNet, ResNet50, and DenseNet201 with the labeled dataset. Then, Cardboard, Organic, and Paper classes are considered biodegradable while the rest are taken as non-biodegradable as shown in figure 1.

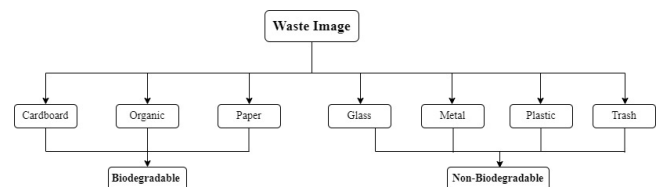


Figure 1: Waste Classification Types

The paper is organized as follows. Section 2 reviews image-based systems for waste classification. Section 3 explains the methodology used. Similarly, section 4 has results and discussion. Finally, Section 5 outlines the conclusions and future work of this study.

2 LITERATURE REVIEW

The authors of [16] proposed a CNN based waste classifier approach for smart and effective waste management. The main objective of

*Corresponding Author: Prakash Poudyal



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this work is to classify waste produced in a household. The authors have used a supervised learning technique and labeled it into four classes namely: paper, cardboard, metal, and plastic. This smart waste classifier used CNN and softmax function with three conv3D layers, 2 hidden layers and 4 output or dense layers. This model gets the best accuracy at 100 epochs with maximum accuracy of 76%. However, this model considers four categories of waste and doesn't consider glass and trash which are general waste in the house. So, by including these categories of data and improving their size, the performance of the model can be improved. The inception-v3 model has been proposed to classify waste images into six categories and gets 92.5% accuracy [3]. In the paper, [18] different CNN-based waste-type classifiers based on VGG-16, ResNet-50, MobileNetV2, and DenseNet-121 have been proposed to classify four types of wastes (general waste, compostable waste, recyclable waste, and hazardous waste). Among them, ResNet-50 waste-type classifier provides an accuracy of 94.86%.

Frost *et al.* [8] developed an image classifier called CompostNet to classify meal waste. The objective of the work is to build a mobile-based application based on an image classification model for correctly identifying compostable, recyclable, and landfill waste. The CNN-based model used "TrashNet" dataset and augmented it by adding 175 and 49 photos of food waste and landfill waste. The authors have built two versions: CompostNet Version A and CompostNet Version B. CompostNet Version A was trained with MobileNetV2 with 4 layers. On the other hand, CompostNet Version B has three convolution layers and breakdown the image into 32 output matrices. Among these two models, Version A has higher accuracy of 77.3%.

In [13], authors use CNN and faster CNN models to classify e-waste and automate the identification of waste equipment size. [4] proposed a waste classification technique using Computer Vision and deep learning. SVM is used for the data preprocessing stage and YOLOv3 (You Only Look Once Version 3) is for waste classification into biodegradable and non-biodegradable waste. Similarly, the paper states that data preprocessing using SVM improves the accuracy of the YOLOv3 algorithm and also reduces memory consumption. Gyawali *et al.* [9] have built different CNN-based waste image classification models namely ResNet50, VGG16, and ResNet18 using Stanford TrashNet dataset into four classes (paper, plastic, metal, and glass). The comparative analysis between these models show that ResNet18 provides 87.8% validation accuracy.

The authors of [2] have fine-tuned TrashNet dataset with different pre-trained model on ImageNet such as DenseNet121, DenseNet169, InceptionResNetV2, MobileNet and Xception. They have also implemented Adam and Adam delta optimizer for training and found better performance on Adam optimizer. Among all those implemented models, DenseNet121 and InceptionResNetV2 performed well with an accuracy of 95% in training and 94% in testing. However, they didn't perform an assessment on each class performance on a model.

3 METHODOLOGY

The most used supervised learning technique in deep learning for image classification is CNN [6]. In this work, we have leveraged a

CNN-based pre-trained model. The architecture of the purposed work has shown in figure 2.

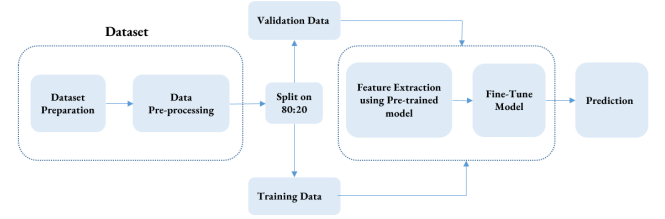


Figure 2: Architecture of purposed work

3.1 Dataset Preparation

The waste images dataset of Stanford TrashNet dataset is used in the experiment [14]. The dataset consists of six classes named cardboard, glass, metal, paper, plastic, and trash. One additional class called "organic" has been added to the dataset from [15] by selecting 612 images randomly. Then, these seven classes have a total of 3242 image data for training and validation. Among them, 103 images have been added from the Internet to the different classes. Similarly, 140 different images have been taken from the Internet for testing a model. The total number of image data is represented in table 1.

Table 1: Number of Images in each class for training and validation

Model	Training Data	Validation Data
Cardboard	322	81
Glass	436	110
Metal	333	84
Organic	489	123
Paper	473	119
Plastic	400	101
Trash	137	35
Total	2590	653

3.2 Transfer Learning

Transfer learning is the improvement of learning in a new task through the transfer of knowledge from a related task that has already been learned or trained [21]. Deep Neural Network required thousands of data for each class as well as it requires high computational power to train the model. So for this work, different pre-trained models such as InceptionV3 [20], InceptionResNetV2 [19], Xception [7], VGG19 [17], MobileNet [11], ResNet50 [10] and DenseNet201 [12] on the ImageNet dataset have been used. These pre-trained model performance metrics have been analyzed properly.

3.3 Image Pre-processing

After allocating train and validation images, pre-processing task have been performed. The images have been resized to 299×299 , 150×150 , 150×150 , 224×224 , 160×160 , 200×200 , and 200×200 for InceptionV3, InceptionResNetV2, Xception, VGG19, MobileNet, ResNet50 and DenseNet201 respectively. Deep learning requires large sets of data for each class. The most popular technique to generate additional images from a small group of an image dataset is to apply a set of transformations to each of them, like rotations, scaling or brightness corrections, zooming, and so on. The transformations for image augmentation used for this work on the training images include rotation and horizontal flipping. Figure 3 and 4 illustrate the sample random images in the dataset and pre-processed images with rotation and horizontal flipping respectively.



Figure 3: Random Images From Dataset With Labels



Figure 4: Pre-processing Image Sample

3.4 Feature Extraction

The procedure is to extract features from new samples learned by a previous network. We used a convolutions layer of a pre-trained network, fed it with new data, and then trained a new classifier on top of it. The features have been extracted from a pre-trained network by freezing or removing the fully connected layers or the last layer. Then, the fully connected layers have been added and trained on the dataset mentioned above.

3.5 Fine Tuning

After feature extraction, fine-tuning is performed, which is a model reuse technique that is equivalent to feature extraction. To execute the categorization, all of the layers save the latest ones were frozen, and a custom layer was created. The global average pooling has

been taken and finally softmax activation function has been used for classification into seven classes. The convolutions layer and the newly incorporated classifier are then trained together, leading to better model performance after fine-tuning.

3.6 Training

Before constructing the model, the dataset was separated into two sets. The training images account for 2590 of the total images, whereas the validation images account for 653. The network is trained using the training images. The validation images are a sample of data from the model's training that is used to measure model competence while tuning the model's hypothesis. The loss criteria are determined using a sparse categorical cross-entropy loss function. Similarly, the Adam optimizer has been used to improve the parameters with a learning rate of 0.0001 in order to obtain the optimal minimum value for 10 epochs for the above-mentioned dataset. Then, the frozen pre-trained have been fine-tuned with the previous results with Adam optimizer with the same learning rate for 10 epochs for pre-trained models excluding the InceptionResNetV2 and DenseNet201 model, these two models are trained for 25 epochs. After the model has been trained, accuracy and loss of training and validation have been noted. Finally, the model weight with higher validation accuracy among trained epochs has been saved. Furthermore, the confusion matrix of the top three performing models has been analyzed and the model with high performance has been evaluated with other metrics like precision, recall, and f1 score.

4 RESULT AND DISCUSSION

Before deciding which model should be utilized in accordance with applicable criteria, the accuracy, confusion matrix, and limitations of different models were compared. InceptionResNetV2, ResNet50, and DenseNet201 were chosen as the final model for the image waste classification.

Table 2: Comparison of Training and Testing Accuracy of Models

Model	Training Accuracy	Validation Accuracy
InceptionV3	0.9872	0.9205
InceptionResNetV2	0.9826	0.9382
Xception	0.9058	0.8933
VGG19	0.6606	0.7181
MobileNet	0.9008	0.8648
ResNet50	0.9849	0.9371
DenseNet201	0.9865	0.9505

Table 2 shows the performance of different models in the same image pre-processing images of the dataset. Among them, the three top-performing models have been DenseNet201, ResNet50, and InceptionResNetV2 based on their performance. So, the performance of those top three models has been explained. The DenseNet201, ResNet50, and InceptionResNetV2 have 0.9865, 9849, and 0.9826 training accuracy with 0.9505, 0.9371, and 9238 validation accuracy respectively. The weight of the highest validation accuracy

has been saved i.e on 23 epochs on the DenseNet201 model. Here, the InceptionV3 model got high training accuracy compared to validation accuracy which indicates quite overfitting in a model. Similarly, DenseNet201 and ResNet50 models have similar training accuracy and less difference in validation accuracy compared to the InceptionResNetV2 model however validation accuracy of DenseNet201 outperforms ResNet50.

Figures 5, 6 and 7 are confusion matrix of ResNet50, InceptionResNetV3 and DenseNet201 model respectively. The InceptionResNetV2 model correctly classifies 16 cardboard, 16 glass, 12 metal, 19 organic, 12 paper, 17 plastic, and 9 trash images out of 140 images having 20 images in each class. Similarly, the ResNet50 model correctly classifies 15 cardboard, 19 glass, 14 metal, 20 organic, 11 paper, 10 plastic, and 6 trash images out of 140 images having 20 images in each class and the DenseNet201 correctly predicted 15 cardboard, 11 glass, 19 metal, 19 organic, 13 paper, 13 plastic, and 12 trash images out of 140 images having 20 images in each class. A total of 101, 102, and 95 images have been classified correctly by InceptionResNetV2, DenseNet201, and ResNet50 out of 140 images which means InceptionResNetV2 and DenseNet201 have almost similar performance on testing data but DenseNet201 have better validation accuracy. The above analysis indicates that DenseNet201 is the most successful model among our trained models.

Table 3: Performance Metrics of DenseNet201 Model

Class	Precision	Recall	F1-Score
Cardboard	0.71	0.75	0.73
Glass	0.1	0.55	0.71
Metal	0.56	0.95	0.70
Organic	0.95	0.95	0.95
Paper	0.65	0.65	0.65
Plastic	0.68	0.65	0.67
Trash	0.80	0.60	0.69

Table 3 illustrates the performance metrics other than training and validation accuracy such as Precision, Recall, and F1-Score of our best-performing model called DenseNet201. These metrics help to understand model performance in each class. The precision calculates the actual correct prediction over the total prediction on a particular class. Similarly, recall computes true positive over true positive and false negative, and the f1-measures provide a weighted average of precision and recall. DenseNet201 model obtained a 0.95 f1-score to the detect organic class whereas the paper class has a low f1-score of 0.65.

Furthermore, we have compared our models with the results of the [2]. The table 4 is the comparison table on validation accuracy with our model which clearly shows our MobileNet and Xception models have performed high validation accuracy than Aral et al. MobileNet and Xception models in 25 epochs. Furthermore, our model has similar accuracy to the top-performing model of [2] as InceptionResNetV2 has 94% validation accuracy however our InceptionResNetV2 model has 93.82% validation accuracy on training 25

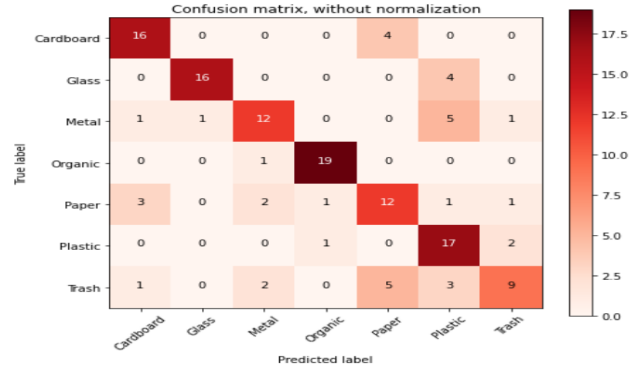


Figure 5: Confusion Matrix of InceptionResNetV2 Model

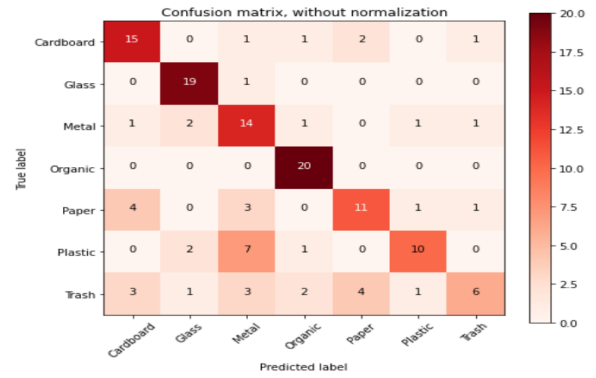


Figure 6: Confusion Matrix of ResNet50 Model

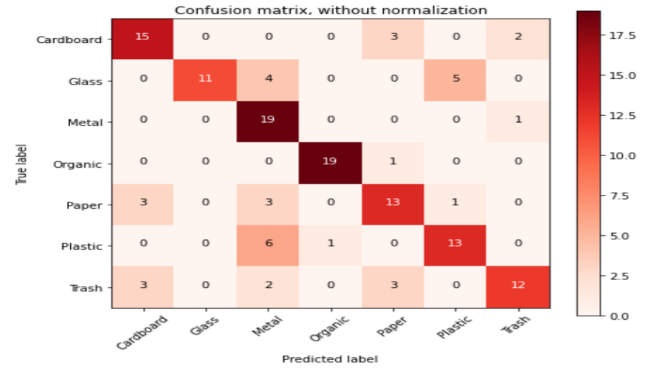


Figure 7: Confusion Matrix of DenseNet201 Model

epochs. Similarly, our DenseNet201 model got 95.19% whereas the DenseNet121 model of [2] acquired 95% validation accuracy.

5 CONCLUSION AND FUTURE WORK

The main focus of this paper has to perform a comparative analysis on pre-trained models like InceptionV3, InceptionResNetV2, Xception, VGG19, MobileNet, ResNet50, and DenseNet201 model with waste image dataset. The transfer learning technique on those seven

Table 4: Comparison with paper [2] on validation accuracy

Model	Paper [2]	Our Model
MobileNet	84% (150 epoch)	86% (20 epoch)
Xception	82% (150 epoch)	89% (20 epoch)
InceptionResNetV2	94% (127 epoch)	93.82% (25 epoch)

pre-trained models has been utilized for waste classification among seven different classes (cardboard, glass, metal, organic, paper, plastic, and trash). After predicting waste images into these classes then cardboard, organic, and paper class images are classified as biodegradable and other classes are classified as non-biodegradable waste. The performance of these trained models with the help of training accuracy, validation accuracy, and confusion matrix are analyzed. Among them, ResNet50, InceptionResNetV2, and DenseNet201 models are the top three performing models. Furthermore, InceptionV3 model got higher training accuracy in comparison with the remaining models however it seems quite overfitting as it has a high variation between validation and training accuracy. However, the InceptionResNetV2, ResNet50, and DenseNet201 models have less variation between training and validation accuracy. Hence, InceptionResNetV2, ResNet50, and DenseNet201 perform better on this dataset.

We used the TrashNet dataset and added an organic class consisting of seven classes and 3242 images to train a model. Therefore, there is the possibility of increasing the performance of the system after appending more number of images for each class and also expanding the classes such as electronic waste, medical waste, and so on. In addition to that, we can increase the epoch size as required while training the model for better learning as well as validation accuracy.

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REFERENCES

- [1] 2016. *3rs - reduce, Reuse and Recycle*. Retrieved May 27, 2022 from <https://www.sustainablesanantonio.com/practices-technology/reduce-reuse-recycle>
- [2] Rahmi Arda Aral, Şeref Recep Keskin, Mahmut Kaya, and Murat Hacıömeroğlu. 2018. Classification of trashnet dataset based on deep learning models. In *2018 IEEE International Conference on Big Data (Big Data)*. IEEE, 2058–2062.
- [3] Fatin Amanina Azis, Hazwani Suhaimi, and Emeroylariffion Abas. 2020. Waste classification using convolutional neural network. In *Proceedings of the 2020 2nd International Conference on Information Technology and Computer Communications*. 9–13. <https://doi.org/10.1145/3417473.3417474>
- [4] Saswati Kumari Behera, L Vasundhara, G Saisudha, NC Haariharan, et al. 2020. AI Based Waste Classifier With Thermo-Rapid Composting. , 4 pages. <https://doi.org/10.1109/ICPECTS49113.2020.9337012>
- [5] Bishal Bharadwaj, Rajesh Kumar Rai, and Mani Nepal. 2020. Sustainable financing for municipal solid waste management in Nepal. *PLOS ONE* 15, 8 (08 2020), 1–15. <https://doi.org/10.1371/journal.pone.0231933>
- [6] K Chauhan and S Ram. 2018. International Journal of Advance Engineering and Research Image Classification with Deep Learning and Comparison between Different Convolutional Neural Network Structures using Tensorflow and Keras. (2018).
- [7] François Chollet. 2017. Xception: Deep learning with depthwise separable convolutions. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 1251–1258.
- [8] Sarah Frost, Bryan Tor, Rakshit Agrawal, and Angus G Forbes. 2019. Compostnet: An image classifier for meal waste. In *2019 IEEE Global Humanitarian Technology Conference (GHTC)*. IEEE, 1–4. <https://doi.org/10.1109/GHTC46095.2019.9033130>
- [9] Dipesh Gyawali, Alok Regmi, Aatish Shakya, Ashish Gautam, and Surendra Shrestha. 2020. Comparative analysis of multiple deep CNN models for waste classification. *arXiv preprint arXiv:2004.02168* (2020). <https://doi.org/10.48550/arXiv.2004.02168>
- [10] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 770–778.
- [11] Andrew G Howard, Menglong Zhu, Bo Chen, Dmitry Kalenichenko, Weijun Wang, Tobias Weyand, Marco Andreetto, and Hartwig Adam. 2017. Mobilenets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861* (2017). <https://doi.org/10.48550/arXiv.1704.04861>
- [12] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q Weinberger. 2017. Densely connected convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 4700–4708.
- [13] Piotr Nowakowski and Teresa Pamula. 2020. Application of deep learning object classifier to improve e-waste collection planning. *Waste Management* 109 (2020), 1–9. <https://doi.org/10.1016/j.wasman.2020.04.041>
- [14] Feyza Ozkefe. 2020. *Garbage CNN* / Kaggle. Retrieved May 27, 2022 from <https://www.kaggle.com/code/rkuo2000/garbage-cnn/data>
- [15] Feyza Ozkefe. 2020. *Waste Classification data* / Kaggle. Retrieved May 27, 2022 from <https://www.kaggle.com/datasets/techsash/waste-classification-data>
- [16] R Sidharth, P Rohit, S Vishagan, R Karthika, and M Ganesan. 2020. Deep learning based smart garbage classifier for effective waste management. In *2020 5th International Conference on Communication and Electronics Systems (ICCES)*. IEEE, 1086–1089. <https://doi.org/10.1109/ICCES48766.2020.9137938>
- [17] Karen Simonyan and Andrew Zisserman. 2014. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556* (2014). <https://doi.org/10.48550/arXiv.1409.1556>
- [18] Chutimet Srinilta and Sivakorn Kanharattanachai. 2019. Municipal solid waste segregation with CNN. In *2019 5th International conference on engineering, applied sciences and technology (ICEAST)*. IEEE, 1–4. <https://doi.org/10.1109/ICEAST.2019.8802522>
- [19] Christian Szegedy, Sergey Ioffe, Vincent Vanhoucke, and Alexander A Alemi. 2017. Inception-v4, inception-resnet and the impact of residual connections on learning. In *Thirty-first AAAI conference on artificial intelligence*.
- [20] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. 2016. Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2818–2826.
- [21] Lisa Torrey and Jude Shavlik. 2010. Transfer learning. In *Handbook of research on machine learning applications and trends: algorithms, methods, and techniques*. IGI global, 242–264.